

IGIUGIG, AK, USA

WMO: 703061

Lat: 59.324N

Lon: 155.902W

Elev: 90

StdP: 14.65

Time Zone: -9.00 (NAK)

Period: 05-19

WBAN: 25521

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-18.8	-30.7	1.0	-24.2	-25.1	1.4	-18.5	38.2	37.2	33.9	32.8	5.1	210	0.775

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	71.8	59.8	67.8	58.1	64.3	56.4	61.9	69.3	59.4	65.5	57.4	62.6	7.0	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
59.1	74.9	63.5	56.5	68.3	61.3	54.8	64.2	59.4	27.6	69.5	25.9	65.8	24.7	62.7	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.7	25.8	21.7	-24.9	78.4	7.6	5.1	-30.4	82.1	-34.8	85.0	-39.1	87.9	-44.7	91.6		
(5)			-24.0	63.1	6.1	3.0	-28.4	65.3	-32.0	67.1	-35.4	68.8	-39.8	71.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.1	14.1	21.3	19.3	32.6	43.2	51.6	55.6	54.7	48.2	36.6	23.0	20.0
	DBStd	18.40	19.97	14.67	13.17	7.88	5.60	4.78	4.56	3.84	5.82	8.75	13.34	15.71
	HDD50	5906	1112	804	951	522	222	33	4	6	95	417	809	931
	HDD65	10918	1577	1224	1416	972	677	402	294	318	503	880	1259	1396
	CDD50	467	0	0	0	0	10	82	178	153	42	2	0	0
	CDD65	3	0	0	0	0	0	1	2	0	0	0	0	0
	CDH74	53	0	0	0	0	2	25	24	2	0	0	0	0
	CDH80	9	0	0	0	0	0	7	2	0	0	0	0	0
	WSAvg	8.5	9.0	8.9	7.8	8.9	9.1	8.3	7.2	7.9	8.3	8.2	9.0	9.5
	PrecAvg	32.5	2.4	2.1	1.9	1.7	1.9	2.2	3.0	3.3	4.3	3.5	3.1	3.1
(15) Precipitation	PrecMax	41.1	6.0	3.8	3.5	2.7	3.9	3.3	4.8	5.6	7.1	6.1	7.2	5.8
	PrecMin	24.6	0.7	0.5	0.4	0.2	0.7	0.7	1.5	1.1	1.8	0.5	0.4	1.4
	PrecStd	4.4	1.1	0.7	1.0	0.6	0.8	0.7	0.8	1.1	1.3	1.2	1.6	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.1	42.7	43.6	52.9	69.4	79.3	77.3	72.0	65.5	55.0	46.6	42.1
		MCWB	39.5	37.9	39.6	42.5	54.2	61.6	62.8	61.2	56.8	49.2	42.8	38.4
	2%	DB	40.8	39.1	40.2	48.4	62.3	70.4	72.4	68.0	61.2	52.7	43.8	40.2
		MCWB	37.6	35.7	36.9	40.8	50.3	57.2	61.1	59.1	54.1	47.8	40.7	37.3
	5%	DB	38.8	37.3	38.0	45.2	56.8	65.1	68.5	64.9	58.2	50.8	41.9	38.8
		MCWB	36.2	34.6	35.3	39.7	46.9	54.9	59.0	57.6	53.4	46.8	39.4	36.2
	10%	DB	37.2	36.0	36.1	42.2	52.7	60.9	64.5	62.0	56.3	48.6	39.5	37.3
		MCWB	34.9	33.7	33.8	38.2	45.3	52.9	57.6	56.4	52.3	45.2	37.2	35.0
	0.4%	WB	39.9	38.2	40.3	43.4	55.3	62.2	66.4	65.1	58.2	50.4	43.7	39.2
		MCDB	43.5	41.5	42.4	49.6	68.8	75.4	72.7	67.5	62.2	53.5	46.3	41.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.9	36.3	37.6	41.7	50.9	58.3	62.7	61.1	56.2	48.8	41.3	37.6
		MCDB	40.5	38.5	39.6	47.4	60.2	67.9	70.2	65.7	59.1	51.8	43.0	39.6
	5%	WB	36.4	35.0	35.4	40.2	48.1	55.9	60.2	58.7	54.5	47.3	39.4	36.4
		MCDB	38.7	36.9	37.3	44.6	54.9	63.7	66.9	63.2	57.3	50.4	41.6	38.5
	10%	WB	34.9	33.8	33.9	38.5	46.1	53.8	58.0	56.9	52.9	45.6	37.4	35.0
		MCDB	37.0	35.6	35.8	42.0	51.6	59.8	63.5	60.8	55.8	48.6	39.5	37.1
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	12.9	15.0	13.5	14.9	15.2	13.0	12.0	13.1	12.0	12.5	11.4
		MCDBR	7.8	10.0	11.7	16.2	24.5	24.2	22.1	18.2	15.7	10.8	9.5	7.9
		MCWBR	6.1	7.8	8.7	10.0	13.1	12.4	11.2	10.0	10.0	7.5	7.8	6.3
	5% WB	MCDBR	7.4	9.5	10.2	14.8	21.0	22.1	20.0	15.3	13.4	9.2	9.7	7.7
		MCWBR	5.7	7.6	7.6	9.3	11.6	11.7	10.7	8.9	9.1	6.7	8.5	6.3
(40) Clear-Sky Solar Irradiance	taub		0.256	0.298	0.312	0.358	0.383	0.379	0.375	0.362	0.330	0.304	0.272	0.256
	taud		2.136	2.203	2.219	2.236	2.274	2.357	2.416	2.450	2.512	2.427	2.108	1.975
	Ebn at Noon		172	225	260	266	268	271	270	265	257	226	157	114
	Edh at Noon		14	23	31	36	38	36	33	30	23	19	15	11
(44) All-Sky Solar Radiation	RadAvg		125	316	829	1254	1521	1667	1400	1125	777	404	168	76
	RadStd		18	58	128	154	165	153	134	133	107	38	25	7

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)	N/A	+5.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page